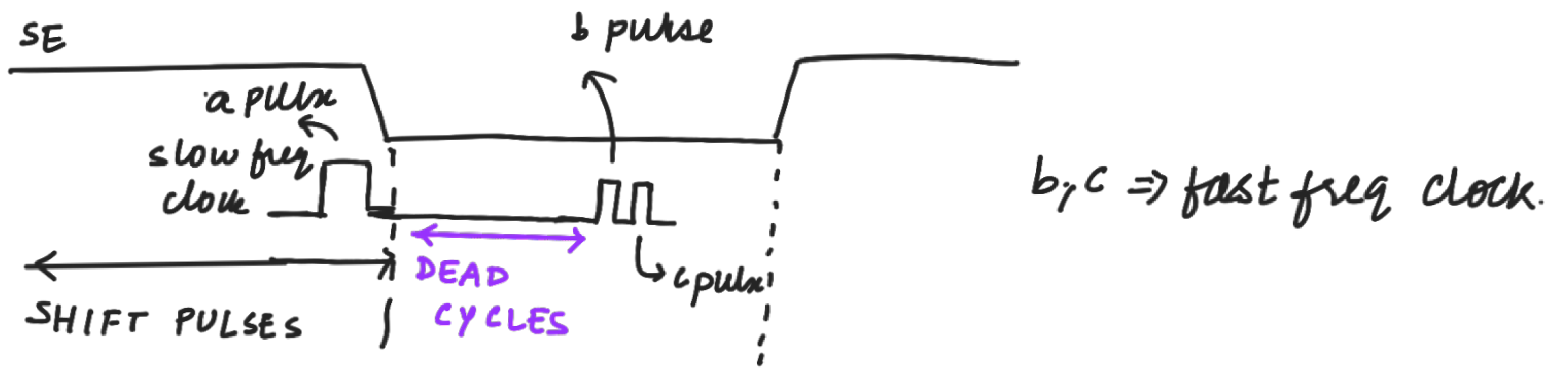




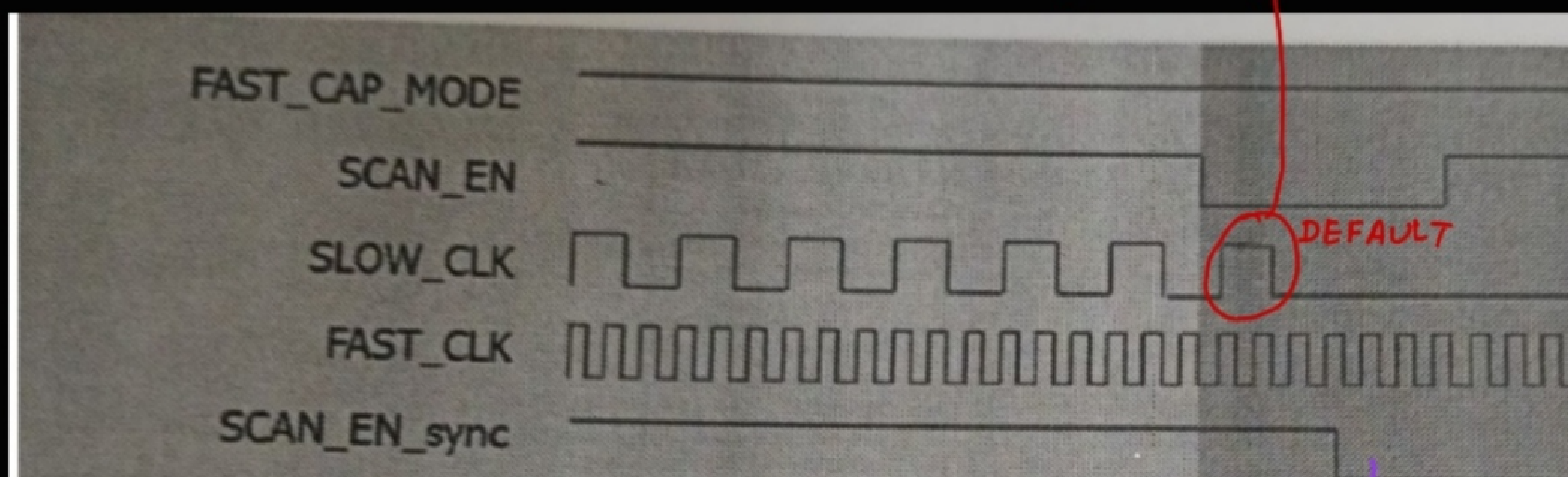
Given to help with the timing closure of the SEN



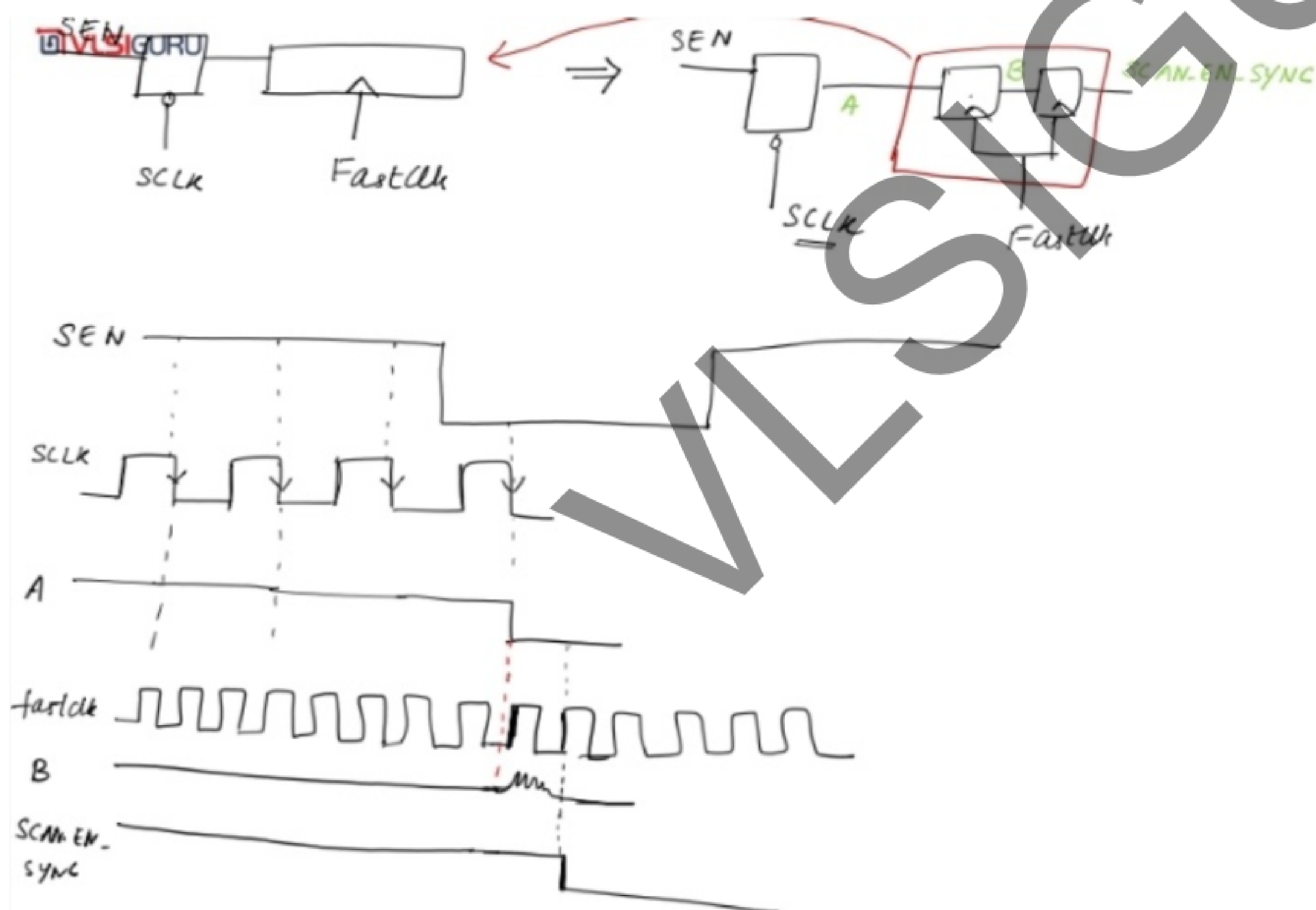
→ scan En changes in the setup time window of b pulse, it will result in metastability.

In OLC,

after this synchronization
of SE will be started



only after scan-en-sync=0, the clock will be given to the oct shift reg's flops which will shift out the data and control the enable signal of ICG in block1. If the enable signal of ICG in block1 is 1, it will allow the clock to propagate to the flops.



BLOCK C :-

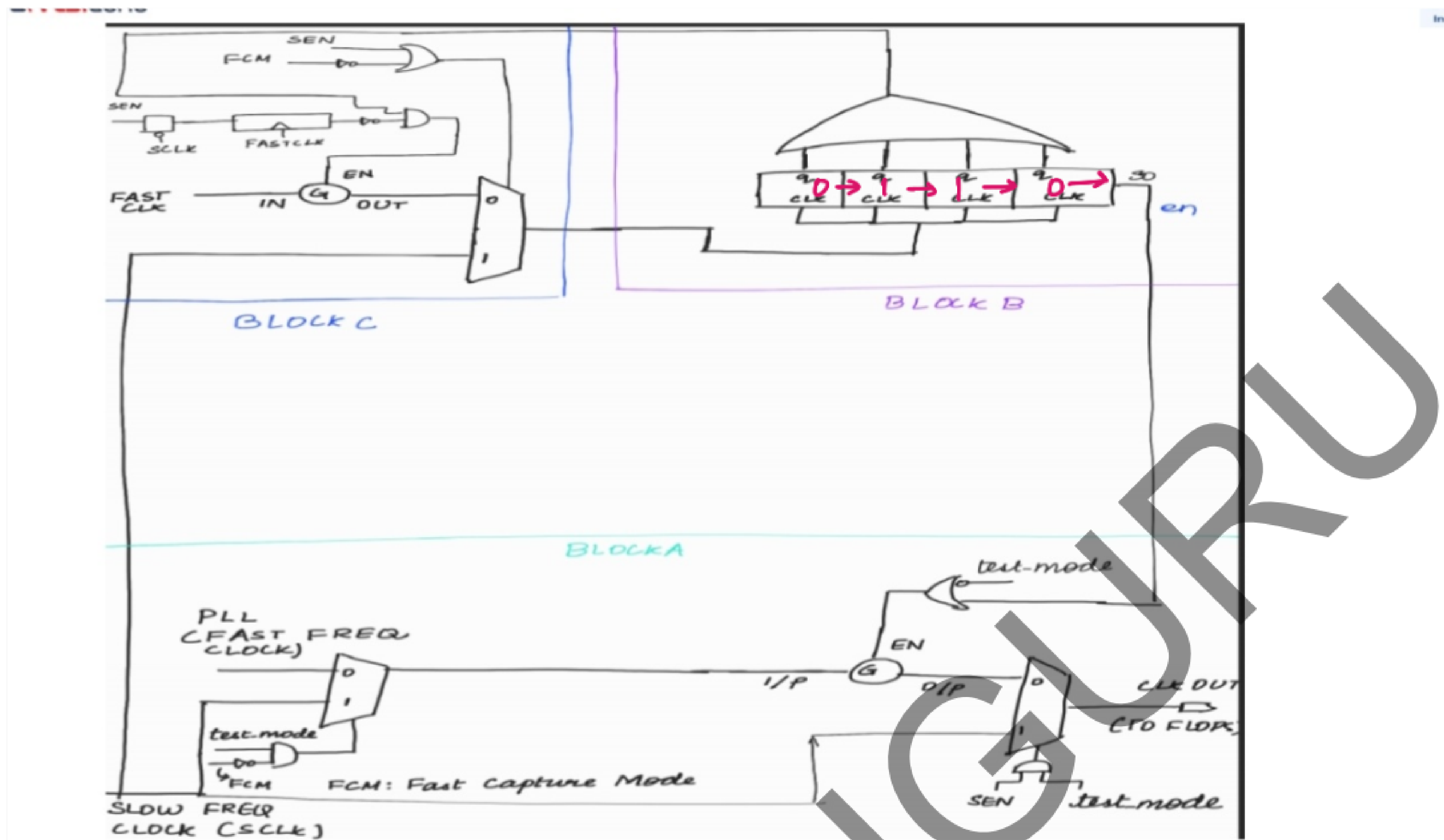
EN CG $\Rightarrow$ AND gate.

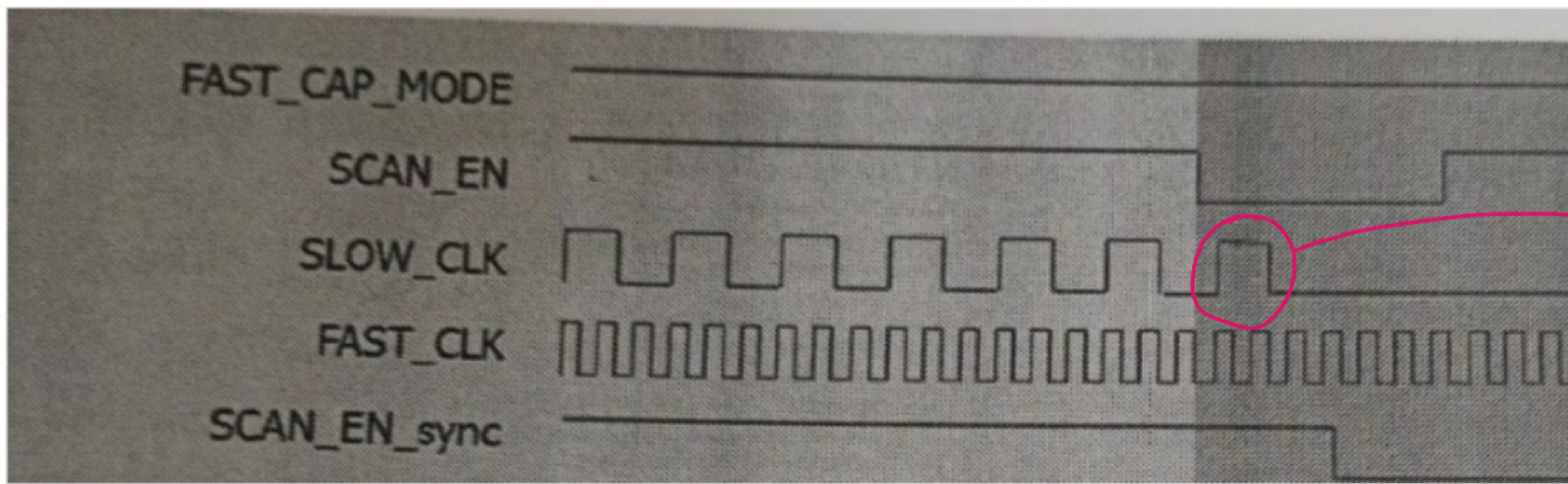
1st IP of AND gate $\Rightarrow$ OR of all outputs in OCC shift registers

2nd IP of AND gate $\Rightarrow$ sync-scan.en

because sync.scan.en = 0

only when SEN is synchronized with fast freq clock domain and when there are some 1s remaining in the OCC shift reg [OR of all $q_s = 1$], clock will be propagated.





→ We can decide when to pulse this clk.
*

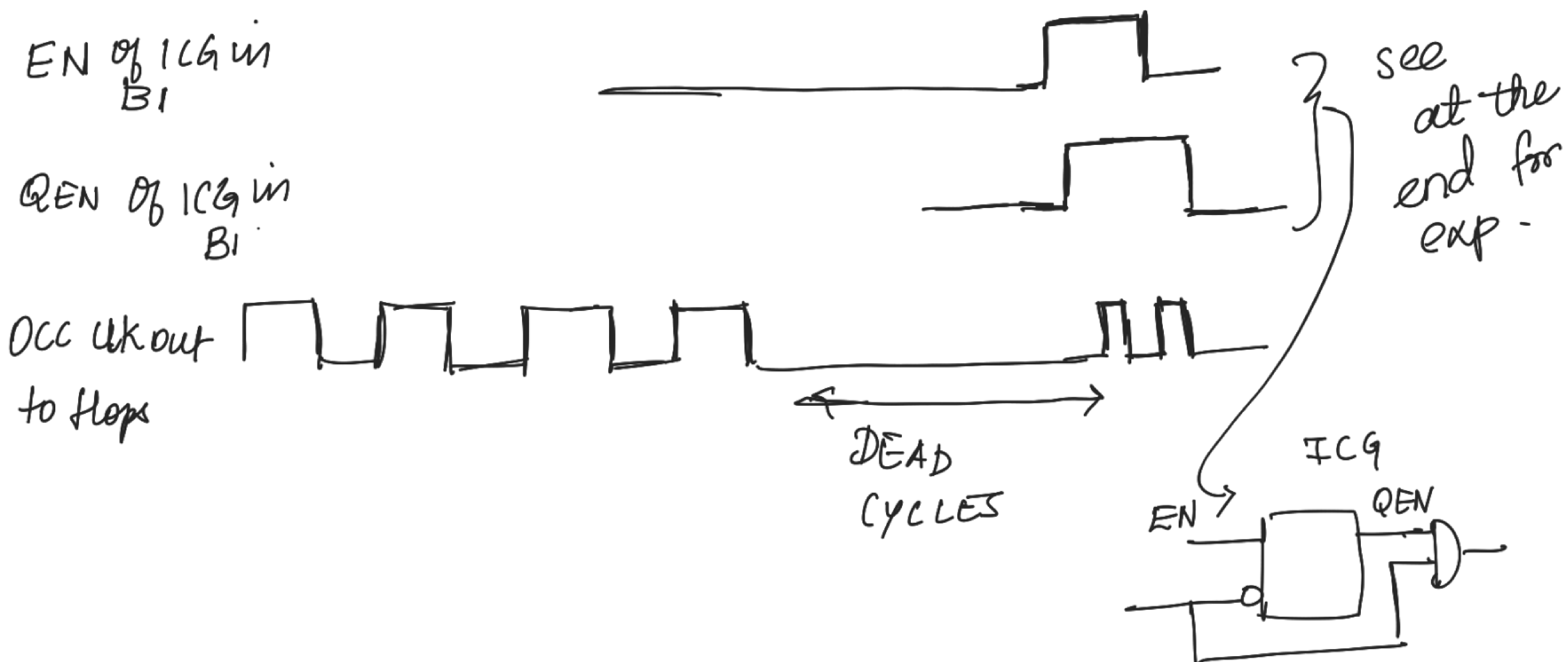
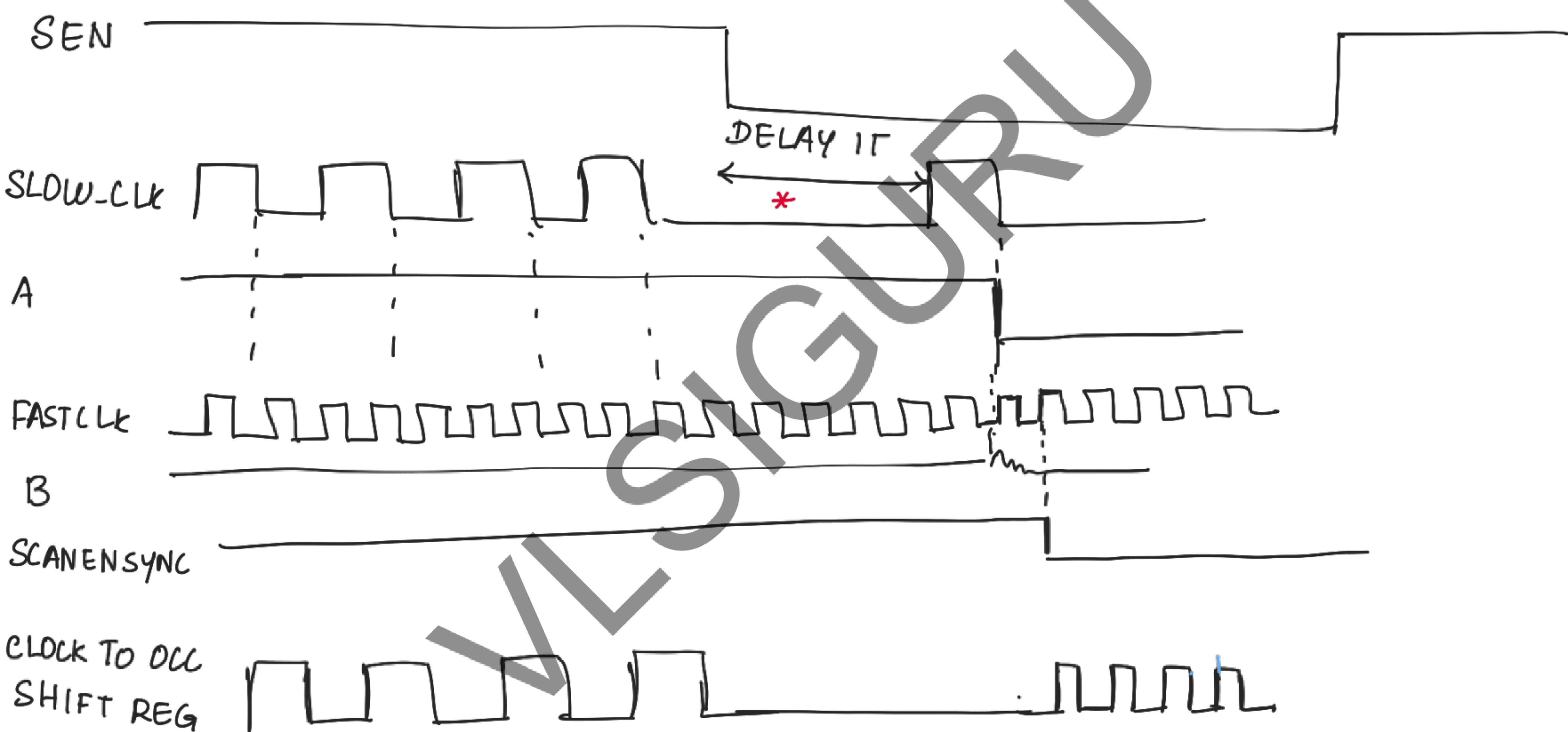
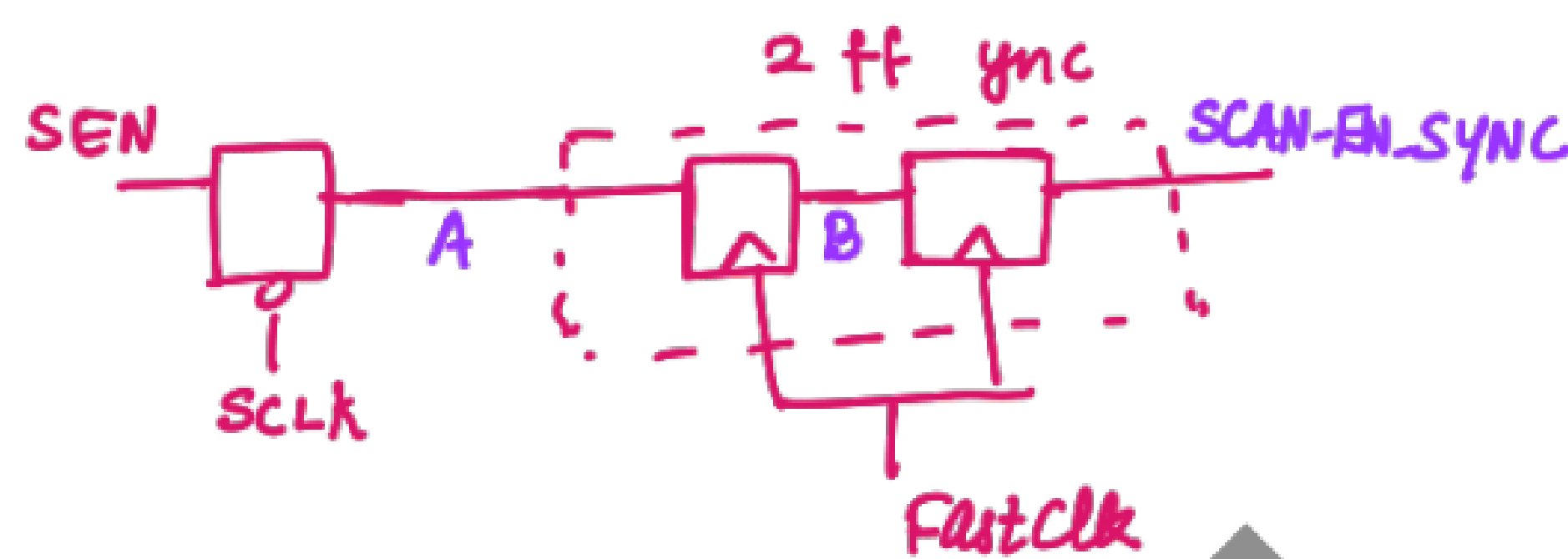
```

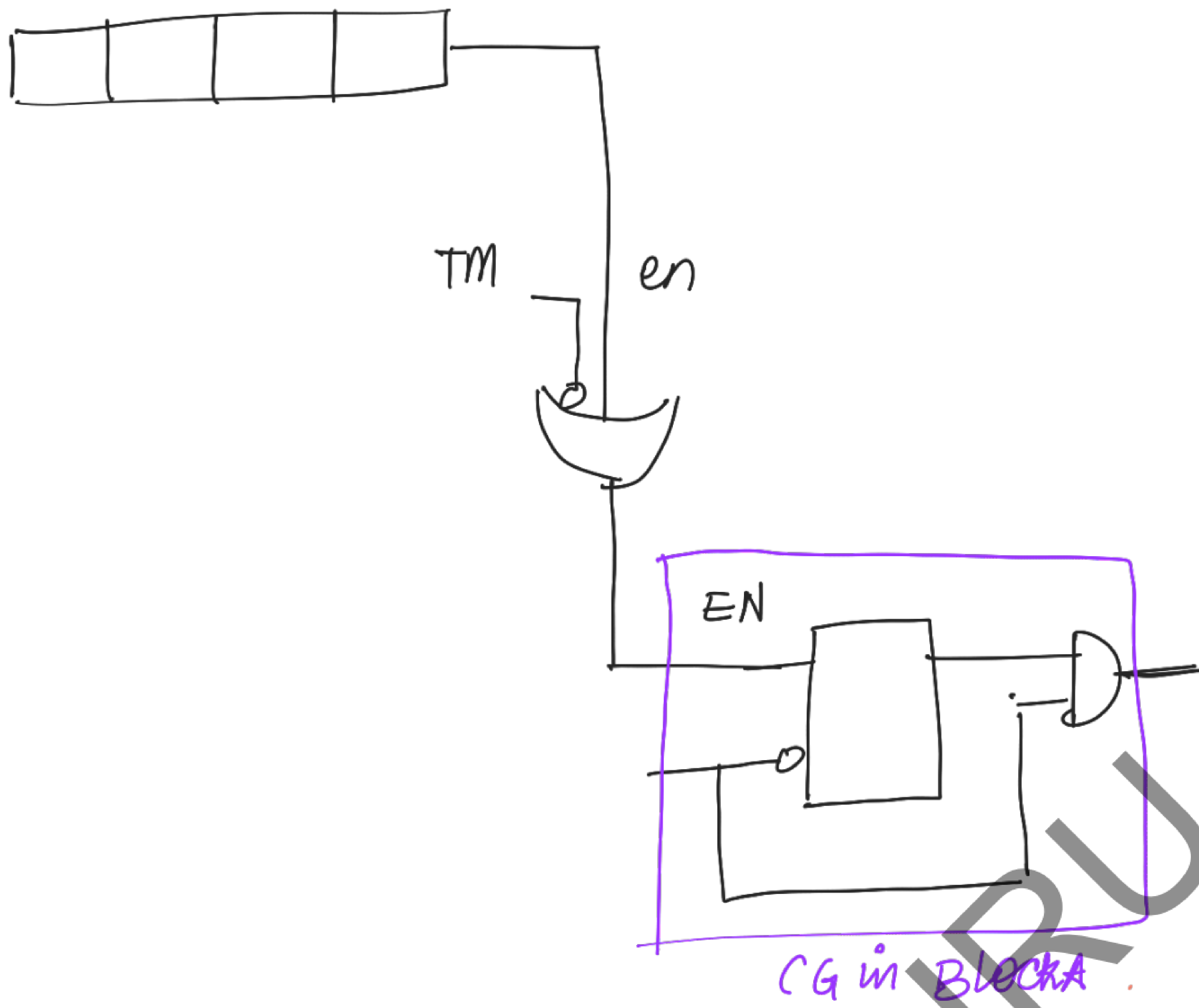
* procedure external_capture ext_fast_cap_proc =
  timeplate gen_tp1;
  cycle =
    force_pi;
  end;
  cycle =
  end;
  cycle =
  end;
  cycle =
  end;
  cycle =
    pulse TEST_CLOCK_top;
  end;
end;

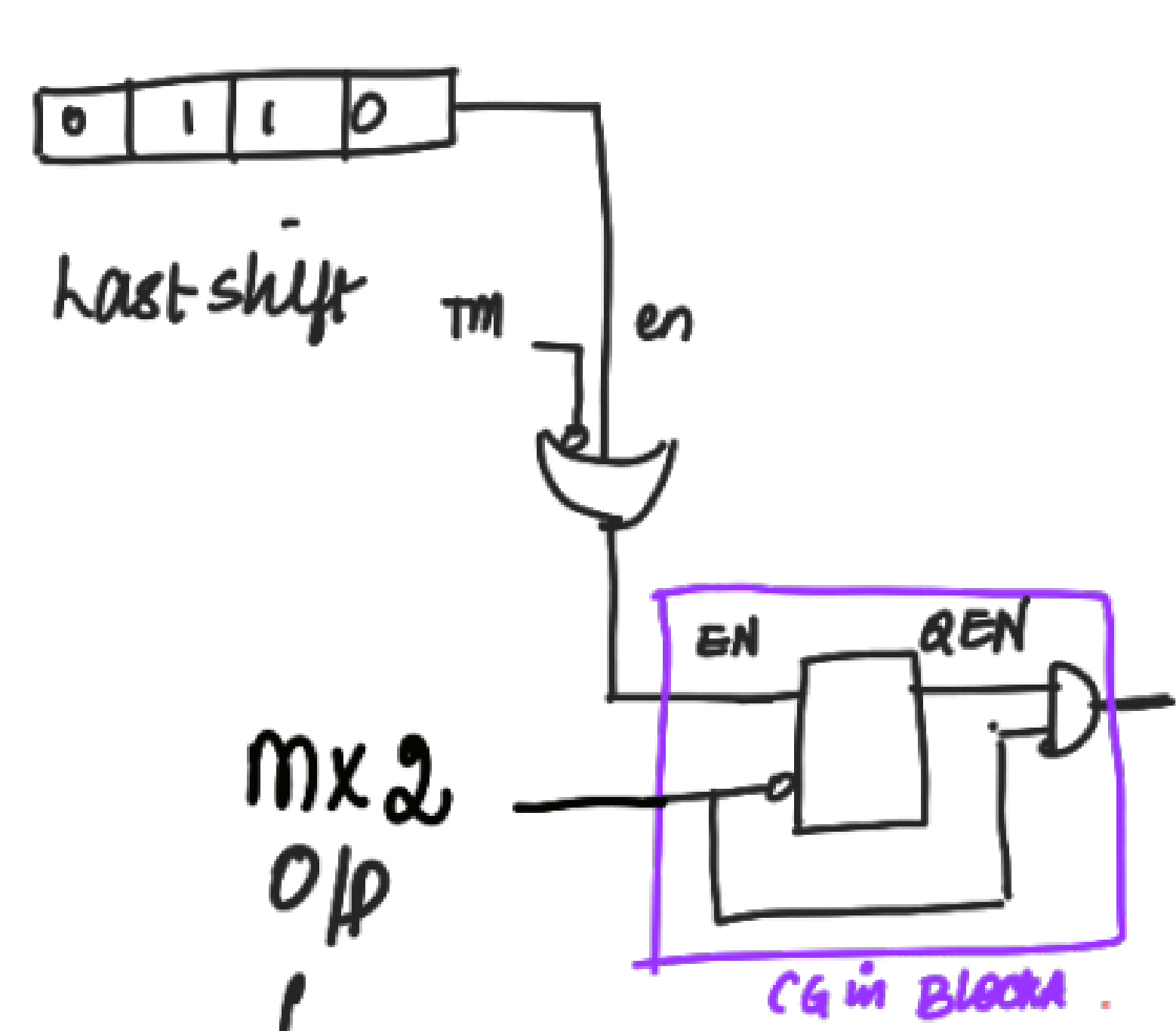
```

3*

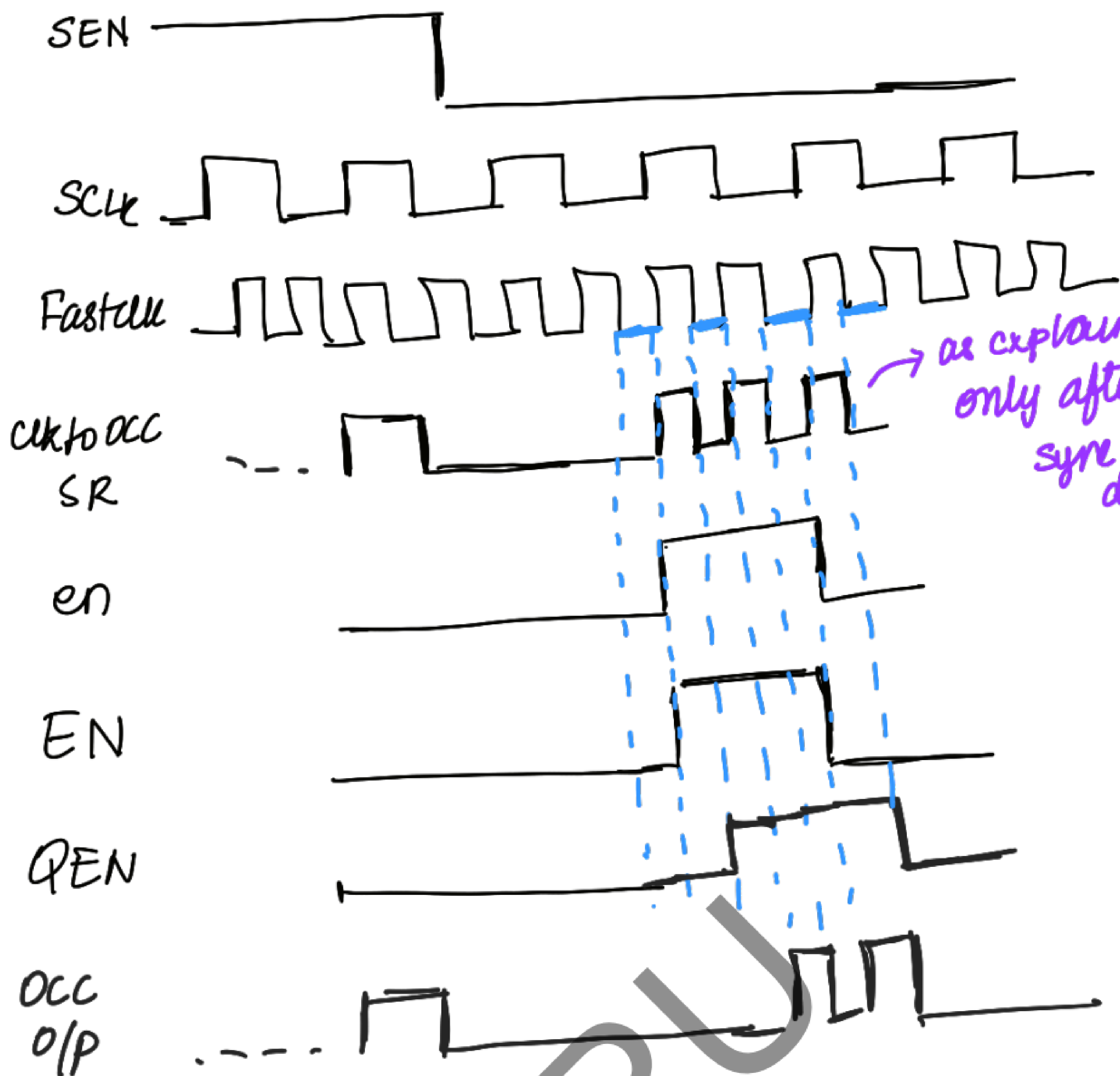
→ in this eg, it is pulsed after 2 cycles.







as discussed
in OCC
topic,
for TDF, it
will be fast clk



as explained above,
only after SEN is
sync with fast clk
domain, clk
will be given
to the OCC
SR ff

VLSIGURU

VLSIGURU